

Neuronal Assembly

The Fundamental Unit of Communication

E pluribus unum. (Out of many, one.)

—GREAT SEAL OF THE UNITED STATES OF AMERICA

United we stand, divided we fall.

—THE SEAL OF KENTUCKY

Success is determined by the collective.

—LÁSZLÓ BARABÁSI¹

The outside-in framework offered an explicit recipe to neuroscience for how to explore the brain: present various stimuli and monitor brain responses. To study learning, we should register the relationship between the stimulus and neuronal response at various stages of learning. Similarly, to study motor behavior, we should establish a reliable relationship between neuronal activity and movement patterns. In principle, anything that the experimenter can conceive, including complex terms that describe various aspects of cognition (check out Figure 1.1 again), can be correlated with neuronal activity.

Recording neurons, one at a time, was one of the first popular techniques in the arsenal of neurophysiology.² The goal for the experimenter is to find the relevant stimulus that can induce reliable neuronal firing. For example, when recording from the ganglion cells of the retina, it would not make sense

1. Barabási (2018).

2. Hubel (1957); Evarts (1964).

to probe their responses with sounds and odorants. Instead, we use visual patterns to probe activity patterns in visual areas. However, as we move deeper into the brain, the idea of a relevant stimulus becomes increasingly more ambiguous. Furthermore, when we study more complex behaviors, such as emotions or memory, the correlational approach needs to be complemented by perturbational methods. Finding the best correlate of the activity of a single neuron and declaring that it responds exclusively or even primarily to, for example, the orientation or color of a visual stimulus, is a heroic venture because the same neurons should be tested over and over, while making sure that the brain state of the animal does not change in the meanwhile. Moreover, finding the physiologically relevant “tuning” response of the neuron may fail if we do not already know its job at least approximately. For example, a neuron may respond relatively selectively to a particular orientation of a black vertical bar even though its main contribution is to color identification. The outside-in approach can provide a tremendous amount of useful information, but it is inherently handicapped because a neuron can be probed only in a few conditions in a given experiment.

Although the spiking activity of a single neuron can offer only limited information, we could perhaps get an average picture of its physiological function by repeating the same experiment again and again, collecting more neurons under the same experimental conditions, and “pretending” that all neurons were recorded simultaneously. This idea is similar to asking each player of an orchestra to play his or her part separately and then combine the individual pieces into a whole. This is the rationale of the single-neuron doctrine.³ A reasonable argument against the single-neuron framework is that, in a flexible brain circuit, neurons can be repurposed for multiple jobs, like a neuroscientist who can also function as a parent, a tennis partner, or a handyman, depending on the circumstances.⁴ More importantly, with single neurons recorded one at a time, one cannot examine how neurons influence each other. A jazz song pieced together from the music played by isolated members will likely not sound right. Based on these considerations, theoreticians began to question that the single neuron is the fundamental unit of computation in the brain. Instead, the idea that a group of neurons that can align themselves for a particular purpose and

3. The classic paper in this field of research is by Barlow (1972).

4. For example, the same neurons in the entorhinal cortex can wear many hats. They may be classified as positional, speed, or head-directional neurons. Although a minority of neurons can appear as specialists, the majority have mixed properties. Instead of individual functional classes, all neurons may have multiplexed properties (Hardcastle et al., 2017). Particular behavioral correlates, designated by the experimenter, may in fact correspond to tails of wide distributions of response features (Chapter 12).

disband themselves when not needed emerged. This is the *cell assembly* or *neuronal ensemble* hypothesis.

CELL ASSEMBLY

The concept of neuronal assembly is one of those things that all neuroscientists talk about, weave into their important findings, and use to explain complex observations—even the workings of the mind—yet it lacks a rigorous definition. As a result, scientists define cell assembly differently. The concept is most often associated with Donald O. Hebb, who coined the term in his classic book *The Organization of Behavior*.⁵ Hebb recognized that a single neuron cannot affect its targets reliably and suggested that a discrete, physically interconnected group of spiking neurons (the cell assembly) is the unit that can represent a distinct percept, cognitive entity, or concept. An assembly of neurons does not live in isolation but communicates effectively with other assemblies. Because of the assumed strong interconnectivity of the assembly members, the activation of a sufficient number of them can activate the entire assembly, a process described in early texts as “ignition” of the assembly. Hebb needed both a large group of interconnected neurons and a trigger because he was in search of “representations” of the outside world and wanted to understand how one signal leads to another.

The assembly idea likely grew out of the Berliner Gestalt psychology at the beginning of the past century, which defined the hypothetical brain substrates of perception on the principles of proximity, similarity, and good continuation.⁶ Hebb’s cell-assembly concept could provide brain-based explanations, at least in principle, for many experimental psychological observations. These

5. I am Donald Hebb’s scientific “grandson.” Cornelius (Case) Vanderwolf was one of his few students and also my postdoctoral advisor (<http://neurotree.org/neurotree/>). Despite our “family” relationship, it is fair to point out that Hebb was not the only person to conceive the assembly or ensemble idea. Similar suggestions were put forward by James (1890), Sherrington (1942), Nikolai Bernstein (1947 in Russian; English translation, 1967), and Konorski (1948). But perhaps the first credit should go to Yves Delage (1919): “every modification engraved in the neuron’s vibratory mode as a result of its co-action with others leaves a trace that is more or less permanent in the vibratory mode resulting from its hereditary structures and from the effects of its previous coactions. Thus, its current vibratory mode reflects the entire history of its previous participations in diverse representations” (Translation by Frégnac et al., 2010). Delage thus had already combined the advantages of oscillations and cell groupings, and his description also relates to Hebb’s second major concept (i.e., the spike timing-based plasticity rule). *Sub sole nihil novi est*.

6. The Gestalt idea of psychology has served as an important and recurring counterpoint to the single-neuron “doctrine” of physiology (Barlow, 1972).

hypothetical neuronal assemblies with assumed discrete boundaries could represent objects or even abstract entities of thought. Coupling two assemblies could serve as a neuronal substrate for associations, as activation of one assembly could then trigger another assembly. Flexibly linking several assemblies into a ring could underlie short-term memory, as reverberation of the activity could persist after the triggering stimulus had vanished. The chaining of such hypothetical cell assemblies, in turn, could support complex cognitive processes, such as memory recall, thinking, and planning, assuming that some internal process can indeed generate such sequences (Chapter 6).⁷

Hebb also realized that a second rule is necessary to flexibly connect assemblies and reinforce these connections so that they can support long-term associations. He suggested a general way that synaptic connections between pairs of neurons could be modified by appropriate timing of their action potentials. Modification of many such connections could form a new memory or an “engram.”⁸ A historical paradox is that his second rule, the Hebbian plasticity rule or simply *Hebb’s rule*, an idea that he borrowed from others, is what earned him a prominent place in the neuroscience hall of fame. In his own words, “The general idea is an old one, that any two cells or systems of cells that are repeatedly active at the same time will tend to become ‘associated,’ so that activity in one facilitates activity in the other.” Hebb’s one-sentence rule became perhaps the most often cited quote in neuroscience, “When one cell repeatedly assists in firing another, the axon of the first cell develops synaptic knobs (or

7. Hebb’s ideas shaped the thinking of numerous researchers who incorporated his ideas in various ways into their brain theories; for example, Miller (1956); Marr (1971); Braitenberg (1971); John (1972); Shaw et al. (1985); Damasio (1989); Abeles (1991); Churchland and Sejnowski (1992); Edelman (1987); Wickelgren (1999); Pulvermüller (2003); McGregor (1993); Miller (1996); Milner (1996); Kelso (1995); Mesulam (1998); Laurent (1999); Varela et al. (2001); Yuste et al. (2005); Harris (2005); Buzsáki (2010) and brain models, e.g., Willshaw et al. (1969); Palm and Aertsen (1986); Hopfield (1982); Amit (1988); Bienenstock (1994); Wennekers et al. (2003). The term “cell assembly” evolved over years, and one might argue that it no longer resembles what Hebb originally intended. In my discussion, I use Hebb’s original definition.

8. The idea of the “engram,” used to define a hypothetical group of neurons representing memories, is closely related to the concept of cell assembly. The term “engram” was introduced by German psychologist Richard Semon (1859–1918; see Schacter, 2001) and popularized by Karl Lashley (1930), who made a lifelong search to identify the engram. Lashley’s “failure” to localize it could be explained by the lack of a good definition or its widely distributed nature. The Nobel Laureate Susumu Tonegawa at MIT has recently revived the term “engram” and performed a series of experiments with optogenetical methods to erase and implant memories into the hippocampus of mice (Tonegawa et al., 2015). For a review of the history and present status of the engram, see Schacter (2001), Josselyn et al. (2015).

enlarges them if they already exist) in contact with the soma of the second cell.” Or, put more simply, “neurons that fire together wire together.”⁹

Hebb’s definition of a cell assembly relies on the structural and physiological connectivity of neurons. Specifically, its members are connected by excitatory synapses created or strengthened by experience. Once a cell assembly is formed, activation of a small group of members can reactivate its entire spatiotemporal signature. The assembly and plasticity concepts of Hebb’s hypothesis defined a program in theoretical neuroscience that has prevailed to the current day. Based on this hypothesis, cognitive psychology established a grand and comprehensive research project to link psychological and physiological processes.¹⁰ It opened the path to tracking down the physiological mechanisms of classification and categorization.¹¹

While the concept of the cell assembly has proved useful, Hebb’s definition made its alleged substrate virtually unidentifiable. In addition, tracking down the signatures of cell assemblies in physiological experiments with the available single-neuron method turned out to be a formidable task. Investigators noted early on the large trial-to-trial variability of neuronal firing patterns. Even the “best” neuron, which robustly responded to a stimulus on some trials, fired one lonely spike or remained completely silent on other trials. Under the cell

9. The original sentence is “neurons wire together if they fire together” (Löwel and Singer, 1992), although the phrase has been most widely popularized Carla Schatz. For demonstration of Hebb’s rule, known also as “spike timing-dependent plasticity” (or STDP) see Magee and Johnston (1997); Markram et al. (1997); Bi and Poo (1998). Prior to Hebb, the timing rule was already well-known from Pavlovian conditioning. The conditional signal (CS) always must precede the unconditional signal (US), and both the CS and US should occur with similar probability to produce successful associations. Reversing the contingency actually makes the CS a predictor of US at a lower than chance probability. Before the in vitro experiments on STDP, Levy and Steward (1983) had already demonstrated the paramount importance of timing in anesthetized rats by associating the contralateral (weak) and ipsilateral (strong, teacher) entorhinal input-evoked local field potential (LFP) responses in the dentate gyrus.

10. The concepts of the cell assembly and spike timing-dependent plasticity have become the guiding rules in many forms of artificial neural networks. Due to connectedness, activity in a few neurons tends to activate all members of an assembly. As a result, the pattern as a whole becomes “auto-associated” and fixed to represent a particular item. The most popular model based on these principles is the *Hopfield attractor network* (Hopfield, 1982). Activity in the Hopfield network varies with time and can jump or move slowly from one stable state (called the “attractor”) to the next. The jumps of the activity bump can be considered to be Hebb’s phase sequence. The number of items (memories) that can be stored with tolerable interference scales with the number of neurons.

11. According to Gerald Edelman, the most fundamental brain operations are integration and segregation (Edelman, 1987; Tononi et al., 1994). Other antonym pairs, such as parsing versus grouping, differentiation versus generalization, and pattern separation versus pattern completion, are also frequently used.

assembly hypothesis, neuroscientists tacitly assumed that the recorded single neurons had several unrecorded partners and that their mean collective behavior was more reliable than the information available from a single neuron.¹² To test the true collective behavior of neurons and prove or dispute Hebb's assembly idea, researchers had to wait until simultaneous recording of large numbers of neurons became possible.

Population Vector

Apostolos Georgopoulos and his colleagues at the University of Minnesota were interested in the neuronal control of movement direction. They trained a monkey to move its arm to one of eight possible targets and observed a striking relationship between the discharge activity of single neurons and the direction of the monkey's arm movement. Many neurons in the motor cortex had a preferred direction of reach. That is, they fired action potentials maximally when the monkey's hand moved in a particular direction, less so when it moved toward a neighboring target, and not at all when the hand moved opposite to the preferred direction (Figure 4.1). Although the experimenters recorded only one neuron at a time, they collected the activity of many neurons, one after the other, and analyzed them as if they had been recorded simultaneously. This cavalier simplification was possible because the monkey's motor task was very stereotypical.

To assess the contribution of all neurons to any given action, Georgopoulos formulated a *population vector* hypothesis. In such a vector, the contributions of neurons with different preferences are summed to produce a final movement command. Each neuron fires the most spikes when the arm moves toward its preferred target, but neurons with nearby preferred directions can also support the same direction less strongly, so the final vote is calculated by vectorial summation of preferred directions of individual neurons weighted by their firing rates. By examining the firing rates of many direction-tuned neurons in a given time window, the population vector model can precisely describe the resulting movement direction. Moreover, when the monkeys were required to translate locations of visual stimuli into spatially shifted reach targets, the population vector accurately predicted mental rotations.¹³ In recent years, using

12. A similar sentiment was also expressed by von Neumann (1958), who suggested that multiple neurons redundantly work together and that their joined activity corresponds to a bit of information.

13. Georgopoulos et al. (1986, 1989). An implicit prediction of the population vector idea is that neurons are relatively equal in their contribution to command muscle movements when

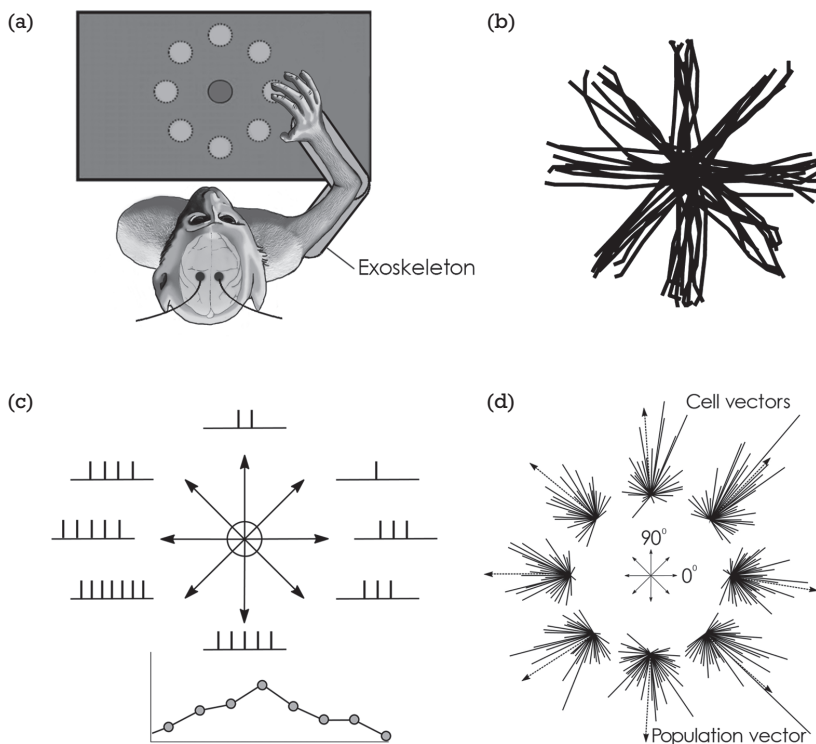


Figure 4.1. Population vector coding of reach movements. A: Monkey reaches to one of eight possible reach directions. B: Trajectory of arm movements. C: Spiking activity of an example neuron. Note very different firing rates in different reach directions. D: Accurate coding of eight movement directions by the population vector computed from the discharge rates of many neurons recorded in the motor cortex. Each one of the eight clusters consists of the weighted preferred directions of the neurons (*solid lines*) and the calculated neuronal population vector (*dotted arrows*). The arrows in the center diagram indicate the direction of the instructed movement. Figure courtesy of Apostolos Georgopoulos.

simultaneous recording of many neurons, improved versions of the population vector method have been used to decode movement direction in real time. It became an indispensable tool in so-called *brain-machine interface* experiments, allowing both monkeys and permanently paralyzed human patients to control

the movement is in the neuron's tuning direction. However, this assumption turned out to be incorrect as the contribution of individual neurons is strongly skewed (Chapter 12). For a contemporary approach to the population coding of arm movement, see Churchland et al. (2012).

computer cursors or robotic arms via thoughts; that is, by the voluntary production of spikes in the appropriate sets of neurons.¹⁴

Relationship Between Population Vector and Cell Assembly

There is a clear parallel between cell assemblies and population vectors. However, in contrast to the mathematically defined population vector, the cell assembly concept is only loosely described as representing “things” through excitatory connections. In line with the prevailing framework of cognitive psychology, the assembly concept has primarily remained a sensory input-driven model of brain activity. The strength of Hebb’s hypothesis is its generalizability, but this is also its major weakness. The idea that changing constellations of neuronal firing patterns underlie our cognitive capacity is not different from the layman’s view and is hardly falsifiable since there is no alternative hypothesis. What makes Hebb’s concept distinct and specific are his suggestions that cell assemblies are formed by experience-driven synaptic changes and that unique assemblies with assumed boundaries can represent separate objects and thoughts.

However, the absence of a rigorous definition raises many questions. Who has more cell assemblies, Albert Einstein, William Shakespeare, you, or me? If cell assemblies are created by training, as is typically the case in artificial neuronal networks, does the brain start with no assemblies and accumulate them over the years? How would the brain’s dynamic landscape look without any experience? What happens to cell assemblies during sleep? How big is a cell assembly? As the average path length in the brain is only a few synapses,¹⁵ every input-triggered ignition would spread to the entire brain through the excitatory connections. Because Hebb’s original assembly concept does not involve inhibition, it is not clear what mechanisms would mark the boundaries among the neuronal groups. Ideally, these questions would be addressed through

14. Using the concept of the population vector, the pioneering Miguel Nicolelis of Duke University and John Donoghue of Brown University have dreamed up a spectacular chapter of translational neuroscience by creating brain–machine interface devices that can read intentions and translate them into the movements of artificial actuators (Nicolelis and Lebedev, 2009; Hochberg et al., 2012). Georgopoulos was a student of the legendary neurophysiologist Vernon Mountcastle, who coined the term “cortical columns” (Mountcastle, 1957). In turn, a prominent student of Georgopoulos, Andrew Schwartz, now at Pittsburgh University, perfected population vector analysis for a more efficient two-dimensional control of a robotic arm in a tetraplegic patient (Collinger et al., 2013). The brain–machine interface field is a rapidly moving translational application of the cell assembly coding ideas.

15. Sporns (2010).

experiments in the learning, behaving animal as opposed to subjective reasoning. However, even when the best tools are available, experiments must be guided by a hypothesis that can be rejected or retained with some confidence.

CELL ASSEMBLIES FOR A PURPOSE

In my view, a fundamental problem with Hebb's "representational concept" is that it presupposes that the same inputs always mobilize the same set of neurons because this framework suggests that objects in the world should correspond to neuronal responses in the brain. We can call it the "neuronal correlate of x " approach. According to the representational outside-in framework, the best strategy by which to understand mechanisms of perception and to identify input-representing cell assemblies is to present various stimuli to the brain and examine the spatiotemporal distribution of the evoked neuronal responses.¹⁶ However, as discussed in Chapters 1 and 3, it is a questionable strategy to identify the neuronal correlates of human-invented mental constructs on the presumption that they must have clear boundaries that correspond to their neural representation. Correlating spiking with external stimulus features is informative to the experimenter, but such correlation is not available to neurons in the brain (Chapter 1). If cell assemblies are meaningful events for neuronal computation, they should have predictable consequences for their downstream partners. Finally, because the brain is also active during sleep, cell assemblies are expected to be as important during sleep as in the waking state (Chapter 8). The outside-in, representational framework is mute about the existence and function of cell assemblies during offline modes of brain operation.

In Chapter 3, I already expressed doubt that the representational strategy can objectively identify cell assemblies because the brain's fundamental priority is not to faithfully "represent" the surrounding world but to simulate practically useful aspects of it based on prior experience and select the most advantageous action in the current situation. From this perspective, an objective definition of the cell assembly requires two related key conditions: a reader classifier and a temporal frame. Let me elaborate on this bold statement a bit. Inspired by the population vector concept, I suggest that a cell assembly can only be defined from the perspective of downstream "reader" mechanisms because the biological relevance of a particular constellation of active neurons (i.e., a presumed cell assembly or assembly sequence) can only be judged from its consequences. In my world, the term "reader" refers to a mechanism that can use the inputs

16. Engel et al. (2001); Hebb (1949); James (1890); Milner (1996); von der Malsburg (1994); Hubel and Wiesel (1962); Rieke et al. (1997).

it receives to respond one way or another. The reader mechanism¹⁷ can be a muscle, a single neuron, groups of neurons, a machine, or even a human observer who interprets the meaning of the inputs. To be meaningful, the same constellation of the assembly should always lead to a similar action, whether it is at the level of a single neuron, neuronal groups, muscles, or hormonal release. The assembly has to have a consistent consequence. This action outcome is what makes any neuronal coalition into a meaningful assembly.¹⁸

Reading the impact of a cell assembly requires a temporal integration mechanism. Neurons come together in time; that is, they synchronize to achieve an action that is not possible for single members alone. Whether events are synchronous or not can be determined only from the point of view of an observer.¹⁹ By extension, I suggest that a neuronal assembly can only be defined from the perspective of a neuronal reader mechanism.

Reader Mechanism–Defined Cell Assembly

All the preceding discussion is based on reasoning and speculation. However, the only way forward for science is by quantitative measurements; in this case, examination of the temporal relationships of spikes among putative assembly members. By the 1990s, my group and other laboratories worked out methods to record sufficiently large ensembles of neurons so that we could address a critical question: What determines the precise timing of a neuron's spikes? We hypothesized, as did Hebb, that the recorded neurons take part in different assemblies, but not all assembly members are active on each occasion. We also

17. Synonyms for “reader” can be terms like “observer,” “classifier,” “integrator,” or “actuator,” depending on whether one’s background is physics, biology, computational science, or engineering. In the present definition, they all refer to the same thing.

18. At the most complex level, such “caused” effects may be plans, memories, decisions, or thoughts (Chapter 13). Berkeley’s dictum “*Esse est percipi*” (“To be is to be perceived”) was a response to British empiricist philosophy. Berkeley famously asked “if a tree falls in a forest and no one is around to hear it, does it make a sound?” (Berkeley, 1710/2010). Galileo Galilei, the most prominent champion of Aristotelian logic, posed a similar question: “if ears, tongues, and noses be taken away, the number, shape, and motion of bodies would remain, but not their tastes, sounds, and odors. The latter, external to the living creature, I believe to be nothing but mere names” (1623/1954). Perhaps nothing is more foreign to contemporary neuroscience than Berkeley’s subjective idealism, yet the idea that in the absence of a “reader-actuator” *meaning* does not exist well resonates with an engineering approach to brain function. Although the metaphor of the reader/interpreter/actuator admittedly has an element of spookiness, it captures the key features of control theory of dynamical systems: a goal or desired output.

19. This is a classic problem in relativity theory (see Chapter 10).

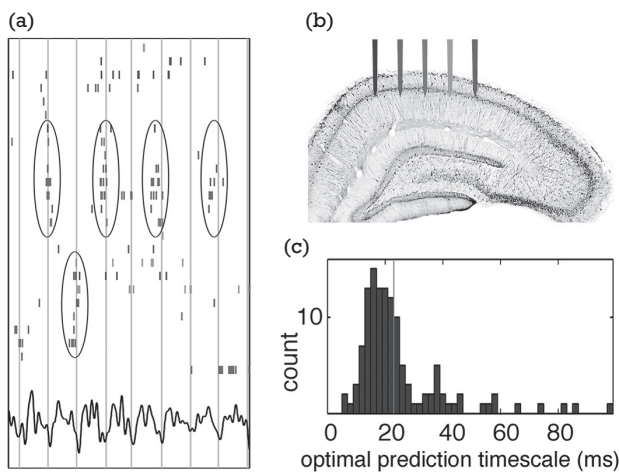


Figure 4.2. Cell assembly: the fundamental unit of neural syntax. A: A raster plot of a subset of hippocampal pyramidal cells that were active during a 1-sec period of spatial exploration on an open field, ordered by their temporal relationship. Each line corresponds to a neuron, and each tick is an action potential in that neuron. Top four ellipsoids indicate a repeatedly active cell assembly, which alternates with another one (*bottom*). Bottom trace shows local field potential dominated by theta and nested gamma oscillatory waves. Vertical lines indicate troughs of theta waves. B: Recording sites from the hippocampal CA1 pyramidal layer in the rat. C: Spike timing of a randomly chosen member neuron of a cell assembly is predictable from peer activity. Distribution of time scales at which peer activity optimally improved spike time prediction of a given cell. The median optimal time scale is 23 ms (*vertical line*).

Modified with permission from Harris et al. (2003).

reasoned that members of the assembly should work together within a measurable time window (Figure 4.2). Like members of an orchestra, neurons in a circuit can effectively time their actions relative to others.²⁰ So we tried to determine the time window within which neurons can best predict the timing of each other's spikes. By varying the analysis window experimentally, we found that the best prediction of the spike timing of single hippocampal neurons from the activity of their peers was when the time window varied between 10 and 30 ms.²¹

This is an important time window in neurophysiology because many physiological variables share it. First and foremost, the membrane time constant (τ) of

20. This project was led by Ken Harris et al. (2003), then a postdoc in my lab. See Truccolo et al. (2010).

21. Jensen and Lisman (1996, 2000); Harris et al. (2003); Harris (2005); Kelemen and Fenton (2010); Lansner (2009).

cortical pyramidal cells is exactly in this range, and it determines their integration ability.²² Discharges of upstream neurons within this window can successfully trigger an action potential in the downstream reader neuron. Generating spike responses in reader neurons is the main reason for a cell assembly to come together. Therefore, from the point of view of a single reader neuron, all neurons whose spiking activity contributes to its own spike can be regarded as a meaningful assembly. Other upstream neurons that fire outside this critical time window (i.e., nonsynchronously) can only be part of another assembly. Thus, by monitoring spiking activity of reader neurons, one can objectively determine whether the upstream neurons are part of the same assembly and serve the same goal (the discharge of the reader neuron) or belong to different assemblies (Figure 4.2). Members of an assembly can project individually to hundreds or thousands of other neurons. Therefore, each neuron can be part of many assemblies, determined largely by its number of targets. These other potential constellations become an assembly when they activate another reader neuron.

In my definition, principal cells do not need to be anatomically connected to each other to form an assembly. It is also inconsequential whether the upstream assembly member neurons are neighbors or reside in multiple corners of the brain. For example, a coactive group of thalamic neurons that lead to the discharge of their target cortical neurons is a meaningful cell assembly. However, by Hebb's definition, these neurons would not be regarded as an assembly because they are not connected to each other by excitatory synapses. In contrast, viewed from the reader neuron definition of the cell assembly, the synchronous activity of these thalamic neurons within the membrane time constant of their downstream cortical neurons renders them an effective assembly of neurons. Therefore, this definition of the cell assembly is closer to the concept of the population vector than to Hebb's definition. However, if the presynaptic neurons are also connected, their synchronous discharge may strengthen their synaptic communication, as predicted by Hebb's plasticity

22. The membrane time constant (τ) is the product of the resistance r_m and capacitance c_m of the membrane ($\tau = r_m c_m$) and represents a duration within which the membrane potential relaxes to approximately 37% ($= e^{-1}$) of its initial value after a voltage step. Neurons with longer τ can integrate postsynaptic potentials for longer time periods (Johnston and Wu, 1995; Koch et al., 1996). The time constant can vary depending on the state of the network in which the neuron is embedded (Destexhe et al., 2003). During intense synaptic activity, such as hippocampal sharp wave ripples, the neuron's input impedance decreases, accompanied by a shortened time window of input integration, corresponding to the fast ripple waves (6–7 ms; Buzsáki, 2015).

rule, and such coalitions may enhance their co-firing in the future.²³ This is a bonus but not a requirement.

Gamma Wave Frame of the Cell Assembly

Another argument for the physiological importance of the cell assembly's ephemeral lifetime is that its 10- to 30-ms time window is similar to the duration of fast synaptic signaling mechanisms. Both excitatory and inhibitory receptors work in this temporal range.²⁴ The temporal interaction between the opposing excitatory and inhibitory postsynaptic effects gives rise to an oscillating tug of war and forms the basis for one of the best known brain rhythms, the *gamma oscillation*.²⁵ The time scale of gamma waves also corresponds to the temporal window of spike timing-dependent plasticity, a mechanism that can modify the synaptic connections between neurons.²⁶ The temporal similarities among the membrane time constant, fast inhibitory and excitatory signaling, the plasticity window, and the period of gamma oscillations make a strong case for the potential importance of upstream cell assemblies acting within this time window, which we discuss further in the next sections.

Cell Assembly Is a Letter in the Brain's Vocabulary

To be effective, cell assemblies acting within single gamma waves (10–30 ms epochs) must mobilize enough peer neurons so that their collective spiking activity can discharge the target (reader) neuron(s). Whether different constellations of spiking upstream neurons are regarded as parts of the same or different assemblies can only be specified by downstream reader neuron(s). Because of the all-or-none spike response of the target neuron, the cell assembly

23. Hebb did not consider inhibitory connections. However, inhibitory neurons interconnecting with excitatory pyramidal cells are very effective in routing spikes in neuronal circuits (Fernández-Ruiz et al., 2017).

24. These are AMPA receptor-mediated excitatory postsynaptic potentials (EPSPs) and GABA_A receptor-mediated inhibitory postsynaptic potentials (IPSPs).

25. Buzsáki et al. (1983); Bragin et al. (1995); Kopell (2000); Whittington et al. (2000); Csicsvari et al. (2003); Bartos et al. (2007); Atallah and Scanziani (2009); Colgin et al. (2009); Buzsáki and Wang (2012); Schomburg et al. (2014); Bastos et al. (2015); Laszotzci and Klausberger (2016). Layer 4 cortical neurons show a resonance and subthreshold oscillations in this time period (Pedroarena and Llinás, 1997), which make them especially sensitive to inputs at this frequency.

26. Magee and Johnston (1997); Markram et al. (1997); Bi and Poo (1998, 2001).

defined by this reader neuron denotes a discrete, collective unitary event, which we can term “fundamental cell assembly”²⁷ or, by analogy to written language, a “neuronal letter.” Several of these gamma assemblies can be concatenated to comprise a neural word (Figure 4.2; Chapter 6).

Although I have written so far from the perspective of a particular reader neuron, in fact, there are no special, dedicated reader neurons. Every neuron can be a reader, and every reader can be part of an assembly, much like members of an orchestra who both produce actions and respond to others’ actions. I simply use the term “reader” as a metaphor to refer to a classifier-actuator mechanism, which sends an action in response to a particular constellation input pattern. Separation of the reader mechanism from the assembly concept is needed only for an objectively quantified definition of neuronal alliances serving well-defined goals. In interconnected local networks, the reader is both an observer-integrator and a contributor in the sense that it generates a measurable and interpretable output. In the simplest case, the output is binary, such as an action potential. In other cases, the reader mechanism may be a population burst or a sequential spiking pattern. Therefore, the reader-centric perspective of assembly organization provides a disciplined framework to uncover the mechanisms that enhance the relationship between upstream firing patterns and the readers of those patterns. This is how the spiking of neurons *becomes* information.

How Big Is a Cell Assembly?

Do neuronal assemblies resemble quartets, chamber orchestras, or philharmonic orchestras? In Hebb’s definition of cell assemblies, membership is defined by connectedness through excitatory synapses. However, as discussed earlier, neither the sufficient nor the total number of assembly members can be determined without first knowing the timeframe and the goal. For me, the question of the Hebb’s assembly size, for example, underlying a visual percept is an unanswerable, ill-posed question. If you catch a glimpse of a couple kissing at a party, the stimulus may transiently activate only a few thousand neurons. On the other hand, if one of the couple is your partner, it may mobilize half of your brain, even though the number of retinal neurons conveying the two sensations is pretty much the same.

27. Again, my definition of the cell assembly is different from other uses. For example, Valentino Braitenberg suggested that, at each moment in time, there is only one assembly active in the entire cortex. His formulation was motivated by a search for the subjective unity of conscious experience. However, to become aware of a sensory experience requires engagement of the appropriate brain networks for at least 500 msec (Libet, 1985, 2005).

Because the reader-centric definition of the cell assembly depends on integrator/actuator mechanisms, it also provides an objective way to approach the question of assembly size. If the goal of an assembly is to discharge a downstream pyramidal cell *in vivo*, the number of neurons whose spikes can be integrated in approximately 20 ms (i.e., in one gamma cycle) can quantitatively define the size of the effective assembly. Approximately 1% of hippocampal pyramidal cells fire in a gamma cycle, and 15,000–30,000 pyramidal cells provide input to each pyramidal cell. Thus, an estimated 150–300 pyramidal cells firing within a gamma cycle may comprise an assembly,²⁸ although these numbers can vary extensively if the synaptic strengths are stronger or weaker. In the olfactory bulb, less than 10% of sharply tuned reader-classifier mitral cells are responsible for generating discrete and defined outputs.²⁹

An inherent difficulty in determining the size of a neuronal assembly is that, without an explicit goal, it is not possible to quantitatively define which neurons belong to the primary assembly and which represent feedback activation of assembly members or newly recruited assemblies serving other goals. Although many neurons can contribute to a cell assembly, the contribution of individual members is not equal, as in an orchestra. The absence of the first violinist is not the same as the absence of a second violinist. Analogously, activity in just a few strongly firing neurons can be as informative about a particular outcome as several dozens of other neurons simultaneously recorded from the same

28. The assembly size, of course, can vary extensively depending on the nature of the integrating reader neuron, synaptic strength, and synchrony of the active upstream population. Under special conditions, when the inputs converge on the same dendritic branch and neurons fire synchronously in less than 6 ms, as few as 20 neurons may be sufficient to initiate a forward-propagating dendritic spike (Losonczy and Magee, 2006). These conditions may be present in the hippocampus during sharp-wave ripples (Csicsvari et al., 2000) and in the geniculocortical system during visual transmission (Wang et al., 2010). If the synapse is strong, a single spike or a burst of spikes in a single presynaptic neuron may be sufficient to discharge a postsynaptic neuron (Csicsvari et al., 1998; Constantinidis and Goldman-Rakic, 2002; Henze et al., 2002; Barthó et al., 2004; Hirabayashi and Miyashita, 2005; Fujisawa et al., 2008; English et al., 2017). In a different approach to estimating the minimum size of a cell assembly to effectively substitute for the effect of a sensory input, channel rhodopsin-2 (ChR2)-expressing neurons in the motor cortex were directly stimulated by light. The mice reported sensing a stimulus when single action potentials were evoked synchronously in approximately 300 neurons. Even fewer neurons (~60) were required when the light induced a train of spikes (Huber et al., 2008). Yet all these considerations will become obsolete when we learn about the strongly skewed distribution of synaptic weights and firing rates in Chapter 12.

29. Niessing and Friedrich (2010). Under special conditions, stimulation of a single pyramidal cell or interneuron can recruit many target neurons in the circuit. Intense trains of intracellularly evoked spikes in a single motor cortex neuron were sufficient to evoke population activity or reset whisking movement in the rat (Miles, 1990; Brecht et al., 2004; Bonifazi et al., 2009; Ellender et al., 2010; Quilichini et al., 2010).

volume. The consequences of such nonegalitarian contribution of neurons to brain function will be discussed in Chapter 12. For now, we can only conclude tentatively that even a small cell assembly in the cortex may involve tens to hundreds of pyramidal cells and their transient partner interneurons, but how these assemblies are concatenated to neuronal words to lead to a percept or an action depends on additional factors.³⁰ Before we discuss the ability of brain circuits to combine neuronal letters to words and words to sentences (Chapters 6 and 7), we need to address the expected advantages of such syntactical operations. Do not skip Chapter 5, please.

SUMMARY

Spikes of single pyramidal neurons are rarely effective in discharging a postsynaptic neuron.³¹ This makes a lot of sense. If a single neuron could discharge its thousands of postsynaptic targets all the time, spikes would have very little value for encoding neuronal information. To effectively send a message, a single neuron must cooperate with its peers. Such cooperation can be achieved by synchronizing their spikes together within the time window limited by

30. We recognize a familiar face in 150 msec or so, and it takes even less time to tell whether we see one or more persons. In this time frame, signals from the retina pass through at least six to eight layers of neurons to the visual cortex, allowing approximately 20 msec processing time for each layer. Knowing the maximum firing rate of neurons (say 100/s), only one or occasionally two spikes can be emitted (vanRullen et al., 2005; Guo et al., 2017) in such a short time (one gamma cycle). In this brief temporal window, the concept of rate coding is meaningless. What matters is how many neurons fire together. From the view of the observing downstream neurons, this togetherness is judged as assembly synchrony.

31. Pioneering studies by Sir Bernard Katz and colleagues established that the strength of a synapse depends on at least three factors: the number of synaptic contacts, the quantal size of the postsynaptic polarization caused by neurotransmitter release from a single synaptic vesicle, and the probability of neurotransmitter release from the presynaptic terminal (Katz, 1966). These parameters vary vastly among the large variety of synapses in the brain. Whereas most cortical pyramidal neurons contact their postsynaptic partners by a single synapse and release a single vesicle (quantum) per spike (Gulyás et al., 1993), some specialized synapses, such as climbing fibers from the inferior olive, make more than 500 synapses onto a Purkinje cell in the cerebellum (Llinás and Sugimori, 1980). The probability of neurotransmitter release is stochastic (Zhang et al., 2009). In the cortex, the probability that a single spike will induce a postsynaptic response is typically very low (0.05–0.5). The probability that a single spike will induce a postsynaptic spike is at least an order of magnitude lower. The distribution of these values across synapses is highly skewed, however (Chapter 12). An important advantage of low release probability is the flexibility it provides. In this way, synapses can act as frequency filters and resonators; they can vary their strength dynamically over both short and longer terms and act as a multiplier/divider gain control mechanism (for a review, see Branco and Staras, 2009).

the ability of the downstream reader neurons to integrate the incoming signals. Therefore, the cell assembly, defined from the point of view of the reader neuron, can be considered a unit of neuronal communication.³²

Acting in assemblies has several advantages. Simple chains of neurons would be vulnerable to synaptic or spike transmission failures from one neuron to the next, resulting in the loss of neuronal messages. Furthermore, minor differences in synaptic weights between the leading and trailing neurons could divert the flow of neuronal traffic in unpredictable ways in the presence of noise. In contrast, cooperative assembly partnership tolerates spike rate variation in individual cells effectively because the total excitatory effect of the assembly is what matters to the reader. Interacting assembly members can compute probabilities, rather than convey deterministic information, and can robustly tolerate noise. We can call the assembly a unit of communication or a putative “neuronal letter.”

32. Spike communication allows neurons to affect other neurons, both near and far, through synaptic release of a neurotransmitter. This is only one of many types of neuronal communication. Nearby neurons can affect each other by electric junctions, ephaptic effects, dendrodendritic release of neurotransmitter, and neuron-glia-neuron effects as well.

